PRANAV SATHEESH

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RESEARCH INTERESTS

Supermassive black hole dynamics and mergers, Galaxy evolution, Gravitational waves, Black hole formation, Cosmological simulations, Multi-messenger astrophysics

EDUCATION

University of Florida 2022 - Present
Ph.D. in Physics
Advisor: Laura Blecha

Indian Institute of Technology Madras 2017 - 2022
BS-MS in Physics

APPOINTMENTS

Research Analyst 2026-2027
Center for Computational Astrophysics
Flatiron Institute, New York

Graduate Research Assistant 2023 - Present
University of Florida
Department of Physics

Graduate Teaching Assistant 2022 - 2023
University of Florida
Department of Physics

PUBLICATIONS

- **Satheesh, P.**, Bhowmick, K. A. & Blecha, L. *Galaxy merger-driven signatures in massive black hole pair hosts across cosmic time*, arXiv:2610.08709, *submitted to ApJ*
- **Satheesh, P.**, Blecha, L., & Kelley, L. Z. (2025). *Mergers and Recoil in Triple Massive Black Hole Systems from Illustris*, ApJ, 993, 222 (2025), arXiv:2506.04369.
- Paul, K., Maurya, A., Henry, Q., Sharma, K., **Satheesh, P.**, Divyajyoti, Kumar, P., & Mishra, C.K. (2024). *ESIGMAHM: An Eccentric, Spinning inspiral-merger-ringdown waveform model with Higher Modes for the detection and characterization of binary black holes*, Phys. Rev. D 111, 084074 (2025), arXiv:2409.13866.

AWARDS

CCA Pre-Doctoral Fellowship by Simons Foundation *2026-2027*
LSST Discovery Alliance (LSST-DA) Data Science Fellowship *2026-2028*
Steigleman fellowship for graduate research in theoretical astrophysics from UF Physics *2024*
Graduate fellowship for first-year graduate students from UF Physics *2022*
62nd Institute day award for academic performance in physics from IIT Madras *2021*
Selected for ThinkSwiss Research Scholarship *2020*
INSPIRE-DST Scholarship for Higher Education *2017 - 2022*

PRESENTATIONS

Invited Talks

- *Massive black hole mergers in cosmological simulations*, **Astrophysics Seminar, New York University**, Sep 2026
- *What can cosmological simulations tell us about massive black hole mergers?*, **Galaxy Coffee Seminar, Max Planck Institute for Astronomy**, July 2026
- *Massive black hole mergers from cosmological simulations*, **Galaxies and Cosmology Seminar, Max Planck Institute for Astrophysics**, July 2026

Conference Talks

- *Massive black hole mergers in cosmological simulations*, **Choir conference on New Data that Challenge Underlying Assumptions in Early Galaxy Evolution, Schoodic Institute** , July 2025
- *Studying gravitational wave and slingshot recoil of massive black holes in cosmological simulations*, **33rd Midwest Relativity meeting, University of Chicago**, November 2023

Posters

- **Satheesh, P.**, Bhowmick, A., Siwek, M., Blecha, L., & Kelley, L. *Massive black hole binaries and mergers from cosmological simulations*, **16th International LISA Symposium** , June 2026
- **Satheesh, P.**, Bhowmick, A., Siwek, M., Blecha, L., & Kelley, L. *Massive black hole mergers and host galaxy properties in cosmological simulations*, **NanoGrav Fall meeting**, November 2025
- **Satheesh, P.**, Blecha, L., & Kelley, L. *Mergers, recoil and slingshot in triple massive black hole systems*, **The era of binary supermassive black holes, Aspen Center for Physics** , February 2025
- **Satheesh, P.**, Blecha, L., & Kelley, L. *Studying gravitational wave and slingshot recoil of massive black holes in cosmological simulations*, **NANOGrav Fall collaboration meeting, University of Michigan**, October 2024
- **Satheesh, P.**, Gandhi, S., & Mishra, C.K. *Fisher analysis of eccentric binaries with higher mode frequency domain inspirals*, **LIGO-Virgo-KAGRA collaboration meeting (virtual)**, March 2022
- **Satheesh, P.**, Saha, P., Schmid, H. *A spectropolarimetric method for predicting the gravitational wave polarization of LISA verification binaries*, **237th American Astronomical Society meeting**, 2021

PROFESSIONAL MEMBERSHIPS

Associate Member, NANOGrav collaboration	*2024* - Present
Graduate Member, American Astronomical Society	*2023 - 2025*
Graduate Member, American Physical Society	*2022-2023*
Member, LIGO Scientific Collaboration	*2021 - 2022*

PROFESSIONAL SERVICE

Referee for **A&A** (2025)

TECHNICAL SKILLS

Programming Languages	Python, C/C++
Software & Tools	Mathematica, LaTeX, Git, GitHub, Shell scripting, hpg
High Performance Computing	Slurm, OpenMP
Python Packages	Pandas, Matplotlib, Numpy, Scipy, Astropy, Scikit-learn
Operating Systems	Linux, Windows, macOS

TEACHING EXPERIENCE

Teaching Assistant, **Code-Astro 2023** Summer 2023
Taught sessions at this software engineering workshop for astronomy supported by the Heising-Simons Foundation.

Teaching Assistant, **Physics-1 with calculus lab** 2022-2023
Introductory laboratory course for undergraduates taking Physics I with calculus at the University of Florida.

Grader, **Complex Networks** Spring 2021
Graduate-level course at IIT Madras.

SCIENCE COMMUNICATION AND OUTREACH

Service

- Organizer, **Comscicon Flagship Workshop** 2025-2026
- Writer and Illustrator, **ComSciWri** 2025 - Present
- Author, **Astrobites** 2023-2025
- Organizer, **Physics Graduate Community** 2023-Present
- Head, **Horizon: The Physics and Astronomy Club of IIT Madras** 2019-2020

Science-communication pieces

- Comic on massive black hole formation 2024
  The Origin Story of Supermassive Black Holes, ClubSciWri
- Science article featured in AAS Nova 2024
  Twinkle twinkle trail of stars how I wonder what you are
- Astrobites articles written by me. 2023-2025

Talks

- *An illustrated guide to supermassive black hole mergers* Feb 2026
  Graduate Student and Postdoc Seminar at UF Physics
- *Scripting the origin story of supermassive black holes* Sep 2024
  Graduate Student and Postdoc Seminar at UF Physics
- *It takes three to tango: Massive black hole triples and the final parsec problem* Sep 2023
  Graduate Student Postdoc Seminar at UF Physics
- *Python for Astronomy*, An Youtube lecture series offered by me as part of Horizon Jul 2020
- *Relativity and Gravitation*, Horizon-IITM Summer School July 2021
- *Analysis of Globular Clusters Using Colour-Magnitude Diagrams*, Shaastra IITM Jan 2020

WORKSHOPS ATTENDED

NANOGrav Astrophysics working group hack week, UC Berkeley *2025*
NANOGrav student workshop, Michigan *2024*
Comscicon flagship workshop, Boston *2024*
ICTS Summer School on Gravitational Wave Astronomy *2023*
School on Black Holes and Gravitational Waves, Chennai, India *2022*